225758: tuffaceous volcanic rock, Wittenoom Gorge

(Dales Gorge Member, Brockman Iron Formation, Hamersley Group, Hamersley Basin, Capricorn Orogen)

Location and sampling

Mount Bruce (SF 50-11), Wittenoom (2553)
MGA Zone 50, 636782E 7530433N

Sampled on 11 September 2017

This sample was collected from outcrop on the east side of a pool on Joffre Creek in Wittenoom Gorge, about 23 km east-southeast of Mount King, 9.2 km north-northeast of Joffre Falls, and 1.4 km southwest of Club Pool.

Geological context

The unit sampled is a tuffaceous volcanic rock within the 2494–2454 Ma Brockman Iron Formation of the Hamersley Group (Martin, 2020a). The Brockman Iron Formation is about 0.5 km thick, consists of banded iron-formation, chert, mudstone, and siltstone, with minor dolerite sills, and conformably overlies the Mount McRae Shale and is conformably overlain by the Weeli Wolli Formation (Martin, 2020a). Numerous tuffaceous horizons within the Brockman Iron Formation have been dated, and indicate depositional ages between 2495 and 2454 Ma (Trendall et al., 1998; Pickard, 2002; Trendall et al., 2004). Tuffaceous horizons within the Dales Gorge Member are marker beds that are commonly referred to as shale bands (Trendall and Blockley, 1968) and abbreviated as DS1–DS16. The volcanic rock sampled at this locality is from DS1 (Fig. 1) near the base of the 2494–2464 Ma Dales Gorge Member, which is 120–180 m thick and the lowermost of four members within the Brockman Iron Formation (Martin, 2020b). An overlying tuffaceous volcanic rock (DS2) from this locality yielded an igneous crystallization age of 2480 ± 7 Ma (GSWA 225760, Wingate et al., 2021c). Volcaniclastic rocks within DS9, sampled about 0.9 km to the southwest, yielded maximum depositional ages of 2477 ± 2 Ma (GSWA 225756, Wingate et al., 2021a) and 2477 ± 4 Ma (GSWA 225757, Wingate et al., 2021b).

Petrographic description

The sample is an altered tuffaceous volcanic rock, consisting of about 95–97% clay minerals, 2–5% chlorite, 1–2% opaque grains, and trace quartz. Clay minerals are very fine grained or cryptocrystalline and iron stained. Oval to elongate features up to 0.1 mm in size consist of moderately birefringent smectite or carbonate minerals. Other oval to equant features are dark brown, up to 0.35 mm long, and composed of coarser-grained, fibrous, iron-stained clay minerals, probably smectite. Platy grains of magnetite or hematite are up to 0.1 mm long, mostly randomly oriented, and rarely weakly aligned. Rare quartz grains are non-abraded and up to 0.05 mm in size. Layering is poorly defined by lenses of extremely fine grained clay minerals or by discontinuous dark folia of iron oxide minerals. The sample is not deformed or metamorphosed.

Zircon morphology

Fifteen zircons isolated from this sample are colourless to pale brown and mainly subhedral. The crystals are up to 150 µm long, and equant to slightly elongate, with aspect ratios up to 3:1. In cathodoluminescence (CL) images, most zircons exhibit concentric zoning. A CL image of all zircons is shown in Figure 2.

Analytical details

This sample was analysed on 18–19 February 2019, using SHRIMP-B. Fifteen analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.12% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.13%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twelve analyses were obtained from 12 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Two analyses are >5% discordant. The dates obtained from these two analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 10 analyses can be divided into two groups, based on their $^{207}Pb^{*}/^{206}Pb^{*}$ ratios.

Group I comprises nine analyses (Table 1), which yield a weighted mean $^{207}Pb^{*}/^{206}Pb^{*}$ date of 2485 ± 9 Ma (MSWD = 2.0).

Figure 1. Outcrop image for sample 225758: tuffaceous volcanic rock, Wittenoom Gorge. The layer sampled for geochronology is indicated

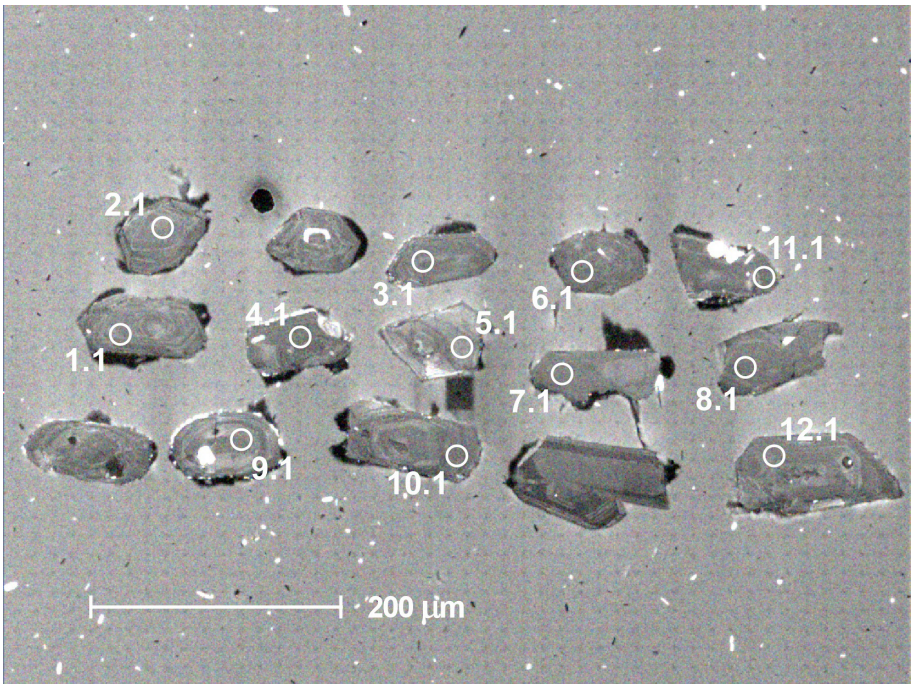

Figure 2. Cathodoluminescence image of all zircons from sample 225758: tuffaceous volcanic rock, Wittenoom Gorge. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 225758: tuffaceous volcanic rock, Wittenoom Gorge

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	6	6.1	451	242	0.54	0.042	2.143	0.015	0.16208	0.00064	2.143	0.015	0.16170	0.00065	2468	15	2474	7	0.2
I	11	11.1	465	310	0.67	0.116	2.176	0.015	0.16285	0.00064	2.179	0.015	0.16182	0.00067	2435	14	2475	7	1.6
I	3	3.1	297	216	0.73	0.033	2.184	0.017	0.16237	0.00076	2.185	0.017	0.16207	0.00078	2430	16	2477	8	1.9
I	1	1.1	195	131	0.67	0.061	2.195	0.019	0.16302	0.00084	2.196	0.019	0.16248	0.00086	2419	18	2482	9	2.5
I	10	10.1	289	199	0.69	0.000	2.142	0.017	0.16299	0.00073	2.142	0.017	0.16299	0.00073	2470	16	2487	8	0.7
I	7	7.1	266	171	0.64	0.020	2.141	0.017	0.16333	0.00077	2.141	0.017	0.16315	0.00077	2470	17	2489	8	0.7
I	5	5.1	164	87	0.53	-0.011	2.114	0.020	0.16327	0.00094	2.113	0.020	0.16337	0.00094	2497	19	2491	10	-0.3
I	9	9.1	194	119	0.61	-0.018	2.172	0.019	0.16375	0.00085	2.171	0.019	0.16391	0.00086	2442	18	2496	9	2.2
I	12	12.1	227	78	0.34	0.000	2.133	0.017	0.16494	0.00079	2.133	0.017	0.16494	0.00079	2478	17	2507	8	1.1
X	8	8.1	322	650	2.02	-0.005	1.997	0.015	0.18099	0.00072	1.996	0.015	0.18103	0.00072	2618	17	2662	7	1.7
D	4	4.1	456	214	0.47	0.081	2.461	0.026	0.15996	0.00065	2.463	0.026	0.15924	0.00067	2197	20	2448	7	10.2
D	2	2.1	260	251	0.97	0.059	2.293	0.019	0.16173	0.00079	2.295	0.019	0.16121	0.00081	2332	16	2468	8	5.5

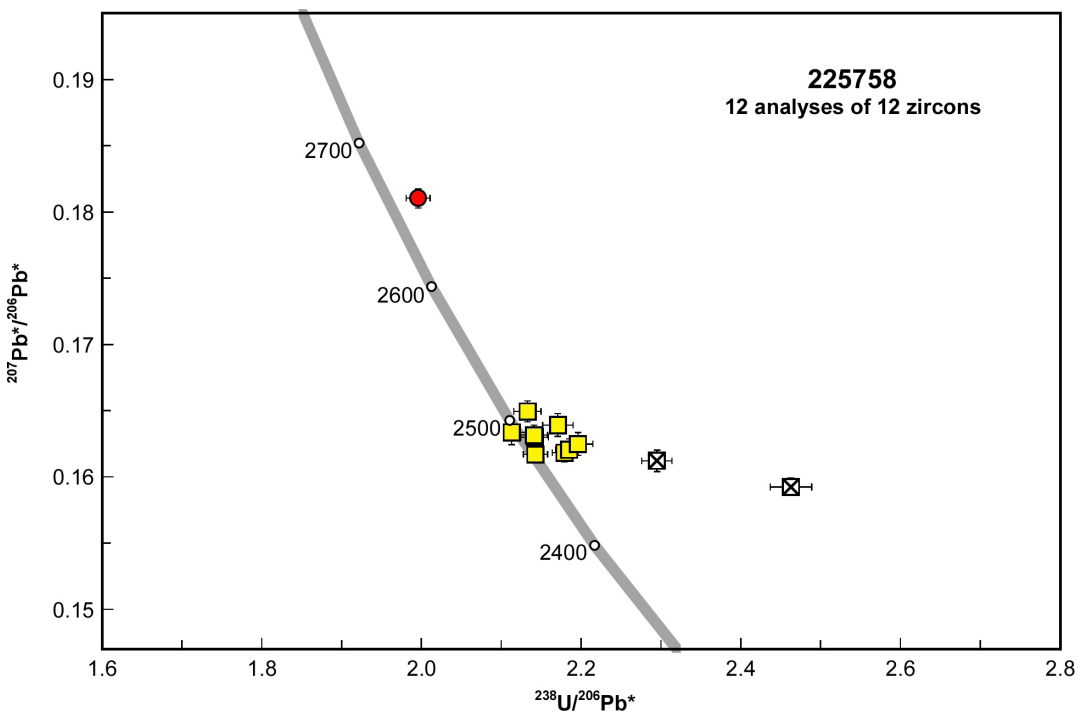

Figure 3. U–Pb analytical data for zircons from sample 225758: tuffaceous volcanic rock, Wittenoom Gorge. Yellow squares indicate Group I (magmatic zircons); red circle indicates Group X (xenocrystic zircon); crossed squares indicate Group D (discordance >5%)

Group X comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2662 ± 7 Ma (1σ).

The date of 2485 ± 9 Ma for the nine analyses in Group I is interpreted as the igneous crystallization age. The date of 2662 ± 7 Ma (1σ) for the single analysis in Group X is interpreted as the age of an inherited zircon.

References

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Recommended reference for this publication

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